Solution of Question Bank

Chapter 1-DFT and FFT 1.

1.Compute the DFT of the 4-pt sequence x(n)= {0,1,2,3}

The four point DFT in the matrix form is given by,

$$X(k) = [W_4] \cdot x(n)$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\therefore X(k) = \begin{bmatrix} 0+1+2+3 \\ 0-j-2+3j \\ 0-1+2-3 \\ 0+j-2-3j \end{bmatrix} = \begin{bmatrix} 6 \\ -2j+2j \\ -2 \\ -2-2j \end{bmatrix}$$

$$\therefore X(k) = \{6, -2+2j, -2, -2-2j\}$$

2.Compute the sequence x(n) for which the DFT is {20, -4+4j, -4, -4-4j}

Since we need to find the sequence x(n), we compute the IDFT.

The IDFT is given by the equation

$$\therefore x(n) = \frac{1}{N}[W_4^*]\,X(k)$$

Since X(k) is of length 4, N = 4 and we generate a 4×4 IDFT matrix.

$$x(n) = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & +1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}\begin{bmatrix} 20 \\ -4+4j \\ -4 \\ -4-4j \end{bmatrix}$$

$$\therefore x(n) = \frac{1}{4}\begin{bmatrix} 8 \\ 16 \\ 24 \\ 32 \end{bmatrix}$$

$$\therefore x(n) = \{2, 4, 6, 8\}$$

Show Calculations :-

$$\boxed{\begin{array}{c}\text{Put} \\ j^2=-1\end{array}}$$

$$x(n) = \frac{1}{4}\begin{bmatrix} 20+4+\cancel{4}j-4-4-\cancel{4}j \\ 20+j(-4+4j)+4-j(-4-4j) \\ 20+4+4j-4-1(-4-4j) \\ 20-j(-4+4j)+4+j(-4-4j) \end{bmatrix}$$

$$\boxed{j^2=-1} = \frac{1}{4}\begin{bmatrix} 20-12 \\ 20+\cancel{4}j+\cancel{4}+\cancel{4}+\cancel{4}j+4 \\ 20+4-\cancel{4}j-\cancel{4}+\cancel{4}+\cancel{4}j \\ 20+\cancel{4}j+4+4-\cancel{4}j+4 \end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix} 8 \\ 16 \\ 24 \\ 32 \end{bmatrix}$$

3.For a discrete time sequence x(n)= {1,2,3,4}, DFT is given by

X(k)= {10, -2+2j, -2, -2-2j}. Compute the DFT of $\hat{x}$(n)= {3,4,1,2} using the circular time shift property of DFT.

Soln. : Given,

$$x(n) = \{1, 2, 3, 4\} \text{ and } \hat{x}(n) = \{3, 4, 1, 2\}.$$

That means $\hat{x}(n)$ is obtained by circularly delaying $x(n)$ by 2 positions. It is as shown in Fig. P. 2.6.5.

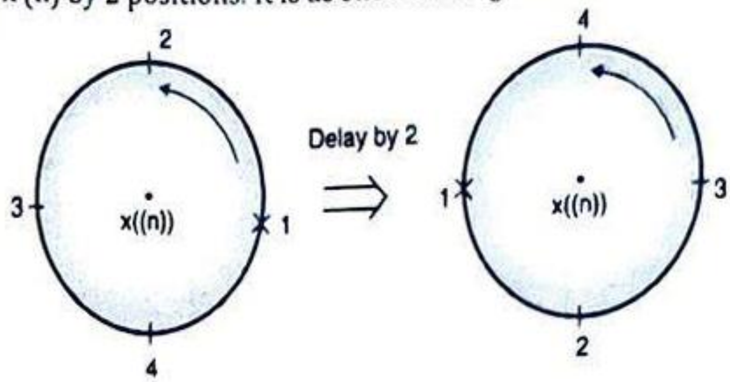


Fig. P. 2.6.5

$$\therefore \hat{x}(n) = x((n-2))$$

If $\quad x(n) \xleftrightarrow{DFT} X(k)$

then $\quad x((n-m))_N \xleftrightarrow{DFT} X(k)\, e^{\frac{-j2\pi mk}{N}}$

Here $\quad m = 2 \text{ and } N = 4$

$$\therefore \hat{x}(n) = x((n-2))_4 \xleftrightarrow{DFT} X(k) \cdot e^{\frac{-j2\pi 2k}{4}}$$

Given, $\quad X(k) = \{10, -2+j2, -2, -2-j2\}$

for $k = 0 \Rightarrow \hat{X}(0) = X(0)\, e^0 = X(0) = 10$

for $k = 1 \Rightarrow \hat{X}(1) = X(1)\, e^{-j\pi}$

$$= (-2+j2)(\cos\pi - j\sin\pi)$$

$$= (-2+j2)(-1) = 2 - j2$$

For $k = 2 \Rightarrow \hat{X}(2) = X(2)\, e^{-j2\pi}$

$$= -2(\cos 2\pi - j\sin 2\pi) = -2$$

for $k = 3 \Rightarrow \hat{X}(3) = X(3)\, e^{-j3\pi}$

$$= (-2-j2)(\cos 3\pi - j\sin 3\pi)$$

$$= (-2-j2)(-1) = 2 + j2$$

$$\therefore \hat{X}(k) = \{10, 2-j2, -2, 2+j2\}$$

4. For an 8-point DFT, the first five DFT coefficients are X(k)= {10,2+3j,1+2j, j,4}. Find the remaining coefficients.

Soln. :

Given :

$$X(0) = 10, \quad X(1) = 2 + 3j, \quad X(2) = 1 + 2j,$$

$$X(3) = j, \quad X(4) = 4$$

We know $\quad X(k) = X^*(N - k)$

In this case $N = 8$

$$\therefore X(5) = X^*(8-5) = X^*(3) = -j$$

$$X(6) = X^*(8-6) = X^*(2) = 1 - 2j$$

$$X(7) = X^*(8-7) = X^*(1) = 2 - 3j$$

Therefore the entire DFT sequence is

$$X(k) = \{10, 2 + 3j, 1 + 2j, j, 4, -j, 1 - 2j, 2 - 3j\}$$

5.Perform circular convolution on the two sequences $x_1(n)$= {1,2,3,4}, $x_2(n)$= {4,1,1,2} using Matrix multiplication method.

Soln. : We generate a circular matrix of $x_2(n)$ and multiply by $x_1(n)$.

$$y(n) = x_1(n) \circledcirc x_2(n)$$

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} 4 & 2 & 1 & 1 \\ 1 & 4 & 2 & 1 \\ 1 & 1 & 4 & 2 \\ 2 & 1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\therefore \text{ We get } \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} 15 \\ 19 \\ 23 \\ 23 \end{bmatrix}$$

$$\text{i.e., } y(n) = \{15, 19, 23, 23\}$$

This is the same result as obtained using the concentric circle method.

Show calculations

$$\begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \end{bmatrix} = \begin{bmatrix} 4+4+3+4 \\ 1+8+6+4 \\ 1+2+12+8 \\ 2+2+3+16 \end{bmatrix}$$

6.If x(n)= {1,2,3,4,5,6,7} and h(n)= {1,0,2}. Perform convolution using the Overlap-Add method.

Soln. : **Overlap-add method :**

In this, the input is divided into blocks of data of size L and (M – 1) zeros are appended to it. This makes the size of data blocks N = L + (M – 1).

In our example we choose a block size of N = 5.

Since h (n) has a length of 3, we append M – 1 = (3 –1) = 2 zeros to L. The length of L = 3.

$N = 3+2 = 5$

$$\therefore \quad x(n) = \{1, 2, 3, 4, 5, 6, 7\}$$

$$x_1(n) = \{1, 2, 3, 0, 0\}$$

$$x_2(n) = \{4, 5, 6, 0, 0\}$$

$$x_3(n) = \{7, 0, 0, 0, 0\}$$

We increase the size of h (n) by appending zeros so that it is equal to N = 5.

$$\therefore \quad h(n) = \{1, 0, 2, 0, 0\}$$

We now circularly convolve each of the blocks with h(n)

$$\therefore \quad y_1(n) = x_1(n) \circledast h(n) = \{1, 2, 5, 4, 6\}$$

$$\therefore \quad y_2(n) = x_1(n) \circledast h(n) = \{4, 5, 14, 10, 12\}$$

$$\therefore \quad y_3(n) = x_3(n) \circledast h(n) = \{7, 0, 14, 0, 0\}$$

Since we had appended 2 zeros to each of the blocks in the beginning, the last 2 terms of each block of the resultants are overlapped and added as shown below,

```
1  2  5  4  6
         ↓  ↓   Add
         4  5  14 10 12
                  ↓  ↓   Add
                  7  0  14  0  0
```

$$\therefore \quad y(n) = \{1, 2, 5, 8, 11, 14, 17, 12, 14\}$$

Show calculations

$y_1(n) = h(n) * x_1(n)$

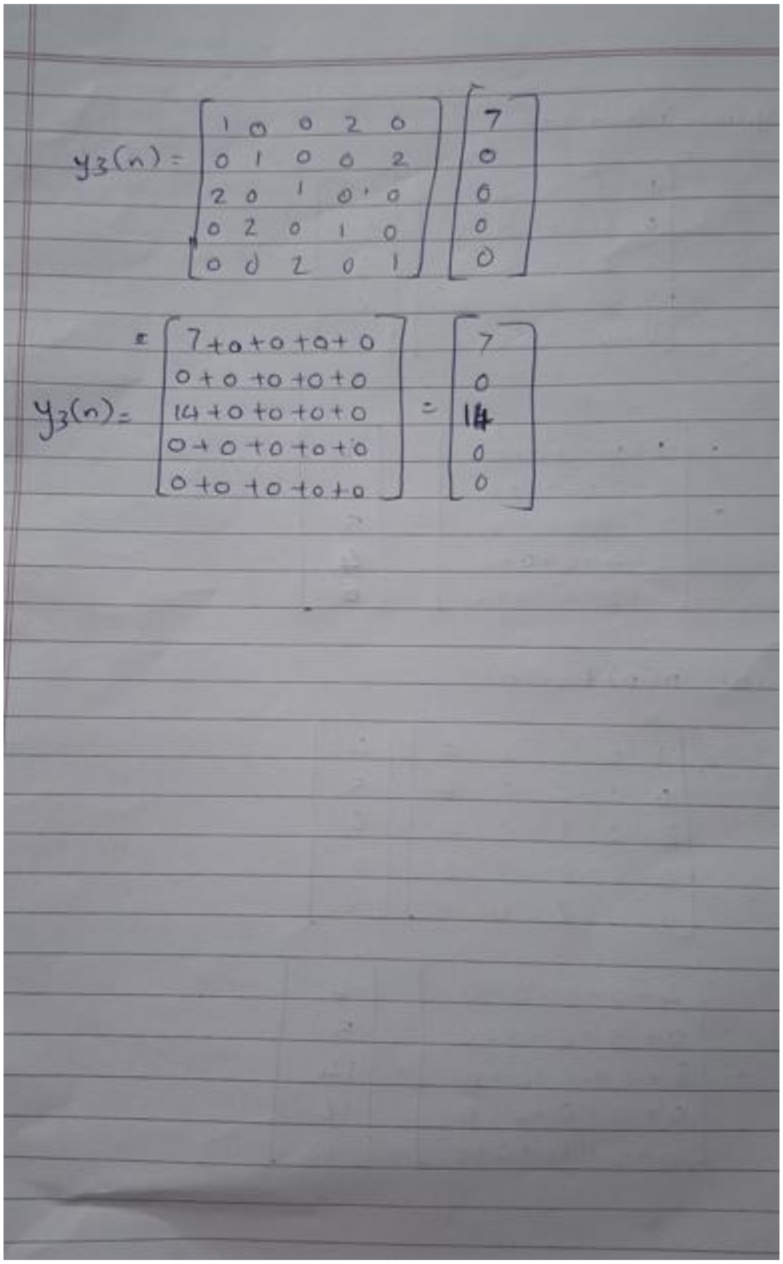
$$= \begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0+0+0+0 \\ 0+2+0+0+0 \\ 2+0+3+0+0 \\ 0+4+0+0+0 \\ 0+0+6+0+0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \\ 4 \\ 6 \end{bmatrix}$$

$y_2(n) = h(n) * x_2(n)$

$$= \begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 4+0+0+0+0 \\ 0+5+0+0+0 \\ 8+0+6+0+0 \\ 0+10+0+0+0 \\ 0+0+12+0+0 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 14 \\ 10 \\ 12 \end{bmatrix}$$



$$y_3(n) = \begin{bmatrix} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$y_3(n) = \begin{bmatrix} 7+0+0+0+0 \\ 0+0+0+0+0 \\ 14+0+0+0+0 \\ 0+0+0+0+0 \\ 0+0+0+0+0 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \\ 14 \\ 0 \\ 0 \end{bmatrix}$$

7. Find the Linear convolution using the Overlap-save method of the following sequences: x1(n)= {1,2, -1,2,3, -2, -3, -1,1,1, 2, -1} h(n)= {1,2,3}

Overlap save method

We will form blocks of length 4.

Here $\quad N = 4$

Length of $h = M = 3$

$\therefore M - 1 = 2$

Now to make each block we need L = 2 data from the previous block. The different blocks are formed as follows :

For the 1st block, we append M – 1 = 2 zeros

$$x_1(n) = \{0, 0, 1, 2\}$$

$$\therefore x_1(n) = \{\underbrace{0, 0}_{\substack{M-1 \\ \text{zeros}}}, \underbrace{1, 2}_{\substack{\text{2 data} \\ \text{from } x(n)}}\}$$

$$x_2(n) = \{\underbrace{1, 2}_{\substack{M-1 \\ \text{from} \\ x(n)}}, \underbrace{-1, 2}_{\substack{\text{New} \\ \text{data}}}\}$$

Similarly other blocks can be formed

$$x_3(n) = \{-1, 2, \underbrace{3, -2}\}$$

$$x_4(n) = \{3, -2, \underbrace{-3, -1}\}$$

$$x_5(n) = \{-3, -1, \underbrace{1, 1}\}$$

$$x_6(n) = \{1, 1, \underbrace{2, -1}\}$$

$$x_7(n) = \{2, -1, 0, 0\}$$

Given $\quad h(n) = \{1, 2, 3\}$

We append zeros to h(n) so that it is equal to N = 4

$$\therefore \quad h(n) = \{1, 2, 3, 0\}$$

We now circularly convolve each of the blocks with h(n)

The different output blocks are calculated as follows :

$$y_1(n) = x_1(n) \circledast h(n)$$

We create a circular of h(n)

$$\therefore \quad y_1(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \\ 1 \\ 4 \end{bmatrix}$$

$$\therefore \quad y_1(n) = \{7, 6, 1, 4\} \qquad \text{....(1)}$$

$$y_2(n) = x_2(n) \circledast h(n)$$

$$\therefore \quad y_2(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 10 \\ 6 \\ 6 \end{bmatrix}$$

$$\therefore \quad y_2(n) = \{2, 10, 6, 6\} \qquad ...(2$$

$$y_3(n) = x_3(n) \circledast h(n)$$

$$\therefore \quad y_3(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \\ 4 \\ 10 \end{bmatrix}$$

$$\therefore \quad y_3(n) = \{4, -6, 4, 10\} \qquad ...(3$$

$$y_4(n) = x_4(n) \circledast h(n)$$

$$\therefore \quad y_4(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} -8 \\ 1 \\ 2 \\ -13 \end{bmatrix}$$

$$\therefore \quad y_4(n) = \{-8, 1, 2, -13\} \qquad ...(4$$

$$y_5(n) = x_5(n) \circledast h(n)$$

$$\therefore \quad y_5(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ -10 \\ 0 \end{bmatrix}$$

$$\therefore \quad y_5(n) = \{2, -4, -10, 0\} \qquad ...(5)$$

$$y_6(n) = x_6(n) \circledast h(n)$$

$$\therefore \quad y_6(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 7 \\ 6 \end{bmatrix}$$

$$\therefore \quad y_6(n) = \{5, 0, 7, 6\} \qquad ...(6)$$

$$y_7(n) = x_7(n) \circledast h(n)$$

$$\therefore \quad y_7(n) = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & 2 & 1 & 0 \\ 0 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 4 \\ -3 \end{bmatrix}$$

$$\therefore \quad y_7(n) = \{2, 2, 4, -3\} \qquad ...(7)$$

The final output y (n) is obtained by discarding initial 2samples from each output block and then by connecting all output blocks one after other.

$$\therefore\ y(n) = \{1, 4, 6, 6, 4, 10, 2, -13, -1, 0, 0, 7, 6, 4, -3\}$$

8.Find 8-pt DFT of the sequence x(n)={1,2,3,4,4,3,2,1} using Radix-2 DIT-FFT.

Note:- For DIT-FFT ⇒ I/P → Bit reversed order
O/P → Normal order

$\{\overset{x(0)}{1}, \overset{x(1)}{2}, \overset{x(2)}{3}, \overset{x(3)}{4}, \overset{x(4)}{4}, \overset{x(5)}{3}, \overset{x(6)}{2}, \overset{x(7)}{1}\}$

1) Find the 8-pt DFT of the sequence $x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$ using Radix-2 DIT-FFT.

Sol^n

Bit revesed input	Stage 1	Stage 2	Stage-3 Output
$x(0) = 1$	$1+4=5$	$5+5=10$	$10+10=20 = X(0)$
$x(4) = 4$	$1-4=-3$	$-3-j=-3-j$	$-3-j+(-1-3j)(\frac{1}{\sqrt2}-j\frac{1}{\sqrt2})$ $= -5.828 - j2.414 = X(1)$
$x(2) = 3$	$3+2=5$	$5-5=0$	$0-0j= 0 = X(2)$
$x(6) = 2$	$3-2=1$	$-3+j=-3+j$	$(-3+j)+(-1+3j)(-\frac{1}{\sqrt2}+j\frac{1}{\sqrt2})$ $= -0.172 - j0.414 = X(3)$
$x(1) = 2$	$2+3=5$	$5+5=10$	$10-10=0 = X(4)$
$x(5) = 3$	$2-3=-1$	$-1-3j$	$(-3-j)-(-1-3j)(-\frac{1}{\sqrt2}-j\frac{1}{\sqrt2})$ $= -0.712 + j4.14 = X(5)$
$x(3) = 4$	$4+1=5$	$5-5=0$	$0+0j=0 = X(6)$
$x(7) = 1$	$4-1=3$	$-1+3j$	$(-3+j)-(-1+3j)(-\frac{1}{\sqrt2}-j\frac{1}{\sqrt2})$ $= -5.828 + j2.414 = X(7)$

$\boxed{j^2=-1}$ Show Calculations

$$-3-j+(-1-3j)\left(\frac{1}{\sqrt2}-j\frac{1}{\sqrt2}\right)$$

$$-3-j-\frac{1}{\sqrt2}+j\frac{1}{\sqrt2}-\frac{3j}{\sqrt2}+j^2\frac{3}{\sqrt2}$$

$$-3-j-\frac{1}{\sqrt2}+j\frac{1}{\sqrt2}-\frac{3j}{\sqrt2}-\frac{3}{\sqrt2} = \left(-3-\frac{1}{\sqrt2}-\frac{3}{\sqrt2}\right)-j\left(1-\frac{1}{\sqrt2}+\frac{3}{\sqrt2}\right)$$

$$=\left(-3-\frac{4}{\sqrt2}\right)-j\left(1+\frac{2}{\sqrt2}\right)$$

$$=-5.828-j2.414$$

$-3+j+(-0.707-j0.707)(-1+3j)$

$-3+j+0.707+j0.707-j2.121+2.121$

$-3+0.707+2.121+j(1+0.707-2.121)$ $\boxed{j^2=-1}$

$-3+j+(-1+3j)\left(-\frac{1}{\sqrt2}+j\frac{1}{\sqrt2}\right)=-3+j+\frac{1}{\sqrt2}-\frac{3j}{\sqrt2}+j\frac{1}{\sqrt2}+j^2\frac{3}{\sqrt2}$

$=-3+j+\frac{1}{\sqrt2}-\frac{3j}{\sqrt2}+j\frac{1}{\sqrt2}+1\times\frac{3}{\sqrt2}$

$=\left(-3+j+\frac{1}{\sqrt2}-\frac{3j}{\sqrt2}+\frac{j}{\sqrt2}+\frac{3}{\sqrt2}\right)$

$=\left(-3+\frac{1}{\sqrt2}+\frac{3}{\sqrt2}\right)+j\left(1-\frac{3}{\sqrt2}+\frac{1}{\sqrt2}\right)$

$=(-3+4/\sqrt2)+j(1-2/\sqrt2)$

$=-0.1715+j\,0.414$

$(-3-j)-(-1-3j)\left(\frac{1}{\sqrt2}-j\frac{1}{\sqrt2}\right)=(-3-j)-\left[-\frac{1}{\sqrt2}-j\frac{3}{\sqrt2}+\frac{j}{\sqrt2}+j^2\frac{3}{\sqrt2}\right)$

$=(-3-j)-\left[-\frac{1}{\sqrt2}-j\frac{3}{\sqrt2}+\frac{j}{\sqrt2}-\frac{3}{\sqrt2}\right)$

$=(-3-j)-\left[-\frac{4}{\sqrt2}-j\left(\frac{3}{\sqrt2}-\frac{1}{\sqrt2}\right)\right]$

$=(-3-j)-\left[-\frac{4}{\sqrt2}-j\left(\frac{2}{\sqrt2}\right)\right]$

$=(-3-j)+\frac{4}{\sqrt2}+j\frac{2}{\sqrt2}$

$=\left(-3+\frac{4}{\sqrt2}\right)-j\left(1-\frac{2}{\sqrt2}\right)$

$=-0.172+j\,0.414$

$$(-3+j)-(-1+3j)\left(-\frac{1}{\sqrt{2}}-j\frac{1}{\sqrt{2}}\right)=(-3+j)-\left(\frac{1}{\sqrt{2}}-j\frac{3}{\sqrt{2}}+j\frac{1}{\sqrt{2}}-j^{2}\frac{3}{\sqrt{2}}\right)$$

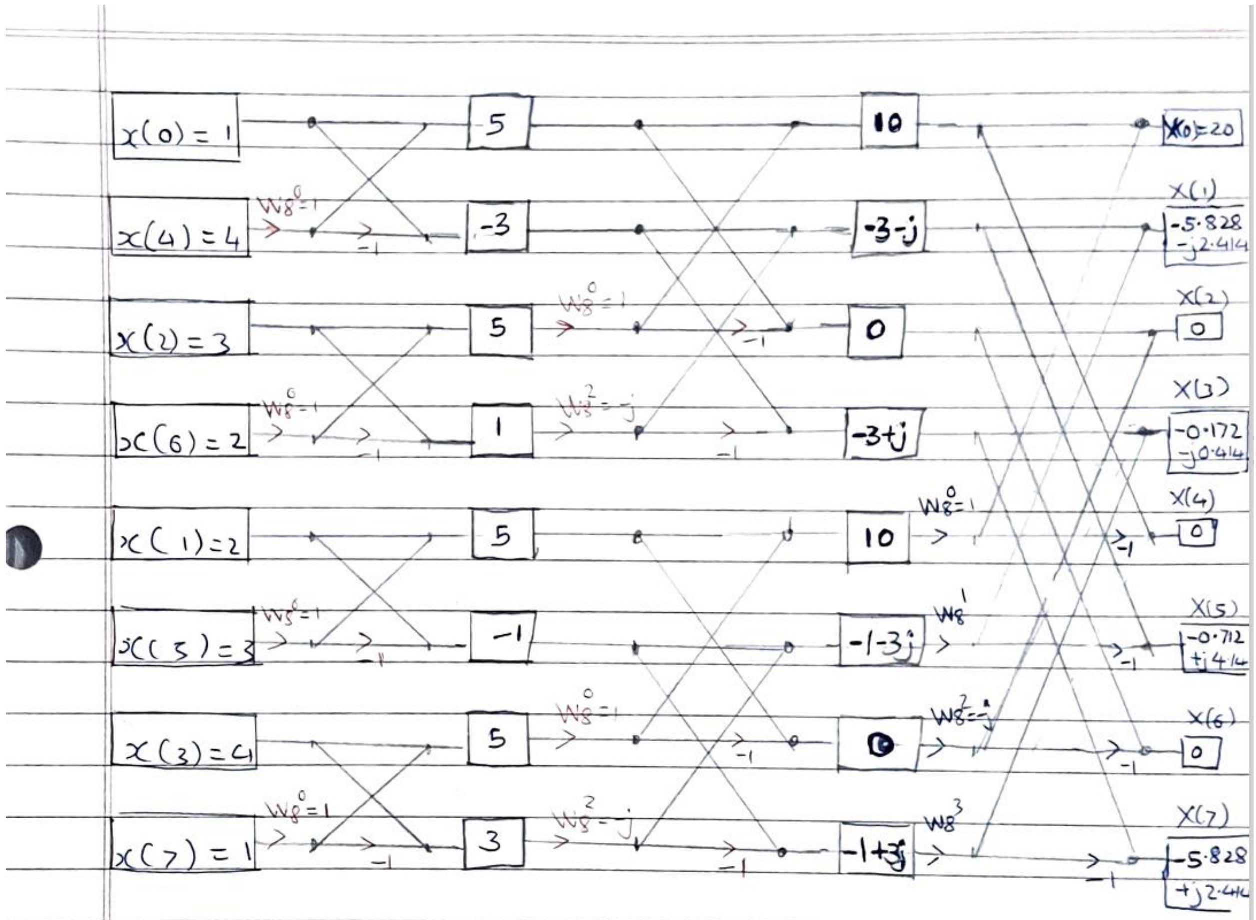
$$=(-3+j)-\left(\frac{1}{\sqrt{2}}-j\frac{3}{\sqrt{2}}+j\frac{1}{\sqrt{2}}+\frac{3}{\sqrt{2}}\right)$$

$$=-3+j-\frac{1}{\sqrt{2}}+j\frac{3}{\sqrt{2}}-j\frac{1}{\sqrt{2}}-\frac{3}{\sqrt{2}}$$

$$=\left(-3-\frac{1}{\sqrt{2}}-\frac{3}{\sqrt{2}}\right)+j\left(1+\frac{3}{\sqrt{2}}-\frac{1}{\sqrt{2}}\right)$$

$$=\left(-3-\frac{4}{\sqrt{2}}\right)+j\left(1+\frac{2}{\sqrt{2}}\right)$$

$$=-5.828+j2.414$$



$W_8^0 = 1,\ W_8^1 = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}},\ W_8^2 = -j,\ W_8^3 = -\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$

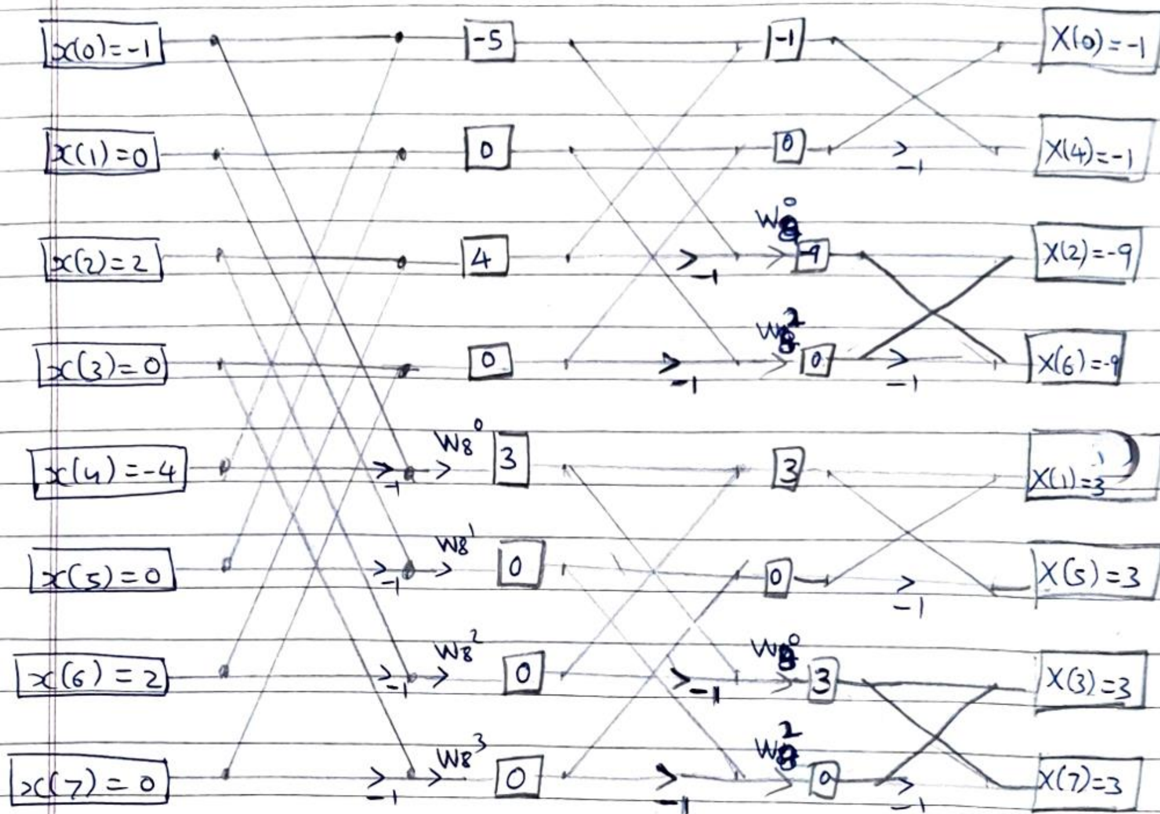
9.Find 8-pt DFT of the sequence x(n)={-1,0,2,0,-4,0,2,0} using Radix-2 DIF-FFT.

Note - For DIF-FFT ⇒ I/P - Normal order
O/P - Bit reversed order

$$x(n) = \{\overset{x(0)}{-1}, \overset{x(1)}{0}, \overset{x(2)}{2}, \overset{x(3)}{0}, \overset{x(4)}{-4}, \overset{x(5)}{0}, \overset{x(6)}{2}, \overset{x(7)}{0}\}$$

Q) Find 8-point DFT of the sequence $x(n) = \{-1, 0, 2, 0, -4, 0, 2, 0\}$ using Radix-2 DIF-FFT.

Input	Stage 1 calculations	Stage-2 calculations	Stage-3 calculations or final output
$x(0) = -1$	$x(0)+x(4) = -1-4 = -5$	$-5+4 = -1$	$-1+0 = -1 = X(0)$
$x(1) = 0$	$x(1)+x(5) = 0+0 = 0$	$0+0 = 0$	$-1-0 = -1 = X(4)$
$x(2) = 2$	$x(2)+x(6) = 2+2 = 4$	$[-5-4]W_8^0 = -9\times 1 = -9$	$-9+0 = -9 = X(2)$
$x(3) = 0$	$x(3)+x(7) = 0+0 = 0$	$[0-0]W_8^2 = 0\times(-j) = 0$	$-9-0 = -9 = X(6)$
$x(4) = -4$	$[x(0)-x(4)]W_8^0 = [-1-(-4)]1 = [-1+4] = 3$	$3+0 = 3$	$3+0 = 3 = X(1)$
$x(5) = 0$	$[x(1)-x(5)]W_8^1 = (0-0)[\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}) = 0[\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}) = 0$	$0+0 = 0$	$3-0 = 3 = X(5)$
$x(6) = 2$	$[x(2)-x(6)]W_8^3 = [2-2](-j) = 0$	$[3-0]W_8^0 = 3$	$3+0 = 3 = X(3)$
$x(7) = 0$	$[x(3)-x(7)]W_8^4 = [0-0]\times(-\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}) = 0$	$[0-0]W_8^2 = 0(-j) = 0$	$3-0 = 3 = X(7)$



$W_8^0 = 1,\ W_8^1 = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}},\ W_8^2 = -j,\ W_8^3 = -\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$

$\therefore$ The DFT is

$X(0) = -1$
$X(1) = 3$
$X(2) = -9$
$X(3) = 3$
$X(4) = -1$
$X(5) = 3$
$X(6) = -9$
$X(7) = 3$

10. How many computations are required to compute a 16-point DFT using the DFT and FFT algorithms?

Here, N = 16

Using DFT

Number of multiplications = $N^2 = 16^2 = 256$

Number of addition = $N(N-1) = 16 \times 15 = 240$

Using FFT

Number of Multiplication = $\frac{N}{2}\log_2 N = 32$

Number of Addition = $N \log_2 N = 64$

11.Compute 4-pt DFT of a sequence x(n)={0,1,2,3} using DIT-FFT and DIF-FFT.

Fig. 1.16.12 shows the DIT-FFT with intermediate values.

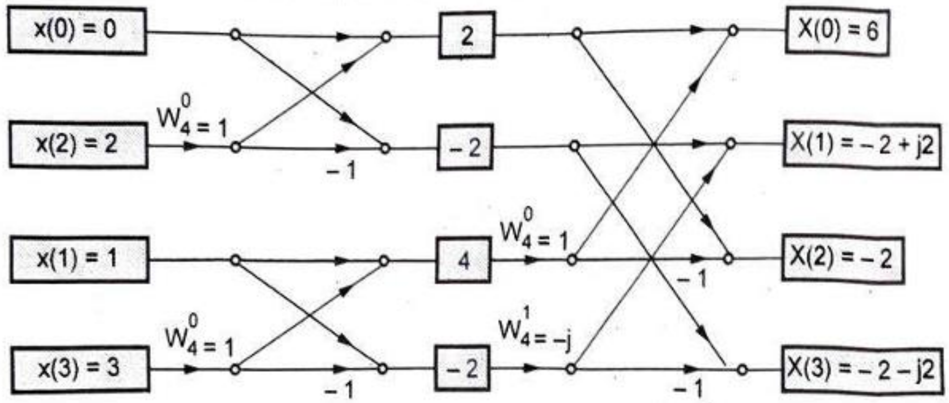


Fig. 1.16.12 DIT-FFT algorithm

DIF-FFT : The twiddle factors remain same. Fig. 1.16.13 shows the DIF-FFT algorithm with intermediate values.

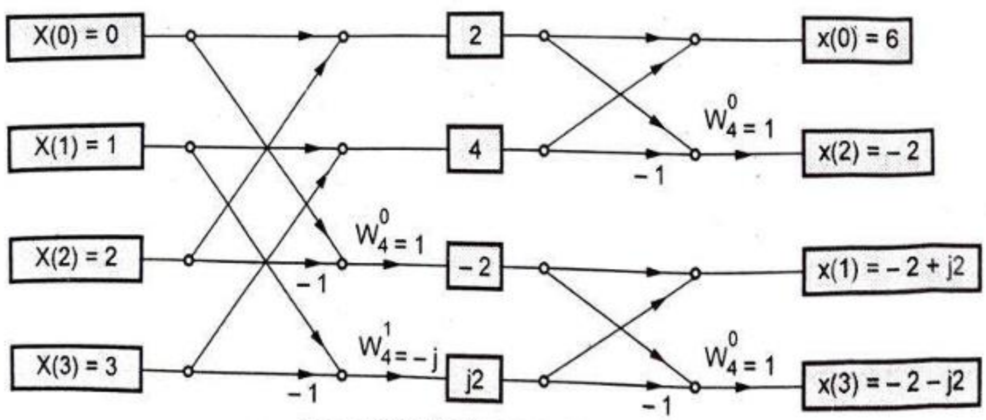


Fig. 1.16.13 DIF-FFT algorithm

From above two signal flow graphs, it is clear that

$$X(k) = \{\underset{\uparrow}{6}, \ -2+j2, \ -2, \ -2-j2\}$$